SonarIQ

Complete UI · design hand-off v1 · class and regulatory compliance checking while you design

Meyer Turku challenge · Since AI 2026
36-hour prototype · tokens: sonariq.css

GOLDEN PATH PROTOTYPE · SECTION 10

Press Play on F0. The panel, the 3D view and the counts move together. Leaving FreeCAD opens the portal.

F0 · P1 — **Idle panel in FreeCAD**
Passenger ship · Deck 5 · 3 packs · 96 checks ready

P2 — **Analyse**
Progress per pack, findings stream in

P3 — **Results**
1 fail · 1 warning · 1 cannot verify · 89 pass · 4 dormant

P4 — **Open F-01**
Evidence: fitted A-0, required A-60, SOLAS II-2/9.2.2.1.1

P5 — **Typed change, Confirm**
“Add four cabins aft of frame 150” → OR-117

P5 — **Ping lands**
14 checks re-run · 3 changed: F-07, F-08, F-09

P5 — **Explanation + 3 fixes**
Each fix already re-checked by the engine

P5 — **Apply fix 2**
One berth per cabin → N_1 64 → 1,340 ≤ 1,350 mm · Pass

P6 — **Accept F-08 with reason**
Goes to the Fire & life safety lead

P7 — **Send to review**
Decisions + open findings + sonariq_rev-b.bcfzip

W2 — **Fire & life safety inbox**
F-07 thread and F-08 waiting for approval

W3 — **Approve F-07**
Signed: name, role, time, revision

PORTAL TRACK

W1 Projects → RCRV card → W4-RCRV

ARTBOARDS · NAMED BY ID

ID	ARTBOARD	PAGE	SIZE
F0	FreeCAD window, panel docked (P1 · P3 · P5)	6	1440 × 900
P1	Idle	7	380 × 860
P2	Running	7	380 × 860
P3	Results	7	380 × 860
P4	Evidence card (F-07; F-01 variant)	7	380 × 860
P5	Chat, golden path as one scroll (unrolled)	7	380 × 860 · 1880
P6	Accept with reason	7	380 × 860
P7	Send to review	7	380 × 860
E1	Edit not allowed	7	380 × 860
E2	Cannot verify (U-01)	7	380 × 860
E3	AI offline	7	380 × 860
E4	Live mode + ping	7	380 × 860
W1	Projects and upload	8	1440 × 900
W2	Discipline inbox	8	1440 × 900
W3	Finding thread (F-07; F-08 variant)	8	1440 × 900
W4	Model viewer · W4-RCRV variant	8	1440 × 900
W5	Import mapping review	8	1440 × 900
W6	Rulebook	8	1440 × 900
W7	Benchmark	8	1440 × 900
W8	Revision impact	8	1440 × 900
C1	Component sheet	9	1440 × 2480
S1	Settings	9	720 × 760
Q1	Meyer review board	9	2840 × 1160

STATUS SET · ICON + WORD, NEVER COLOUR ALONE

Fail · Warning · Pass · Accepted with reason · Cannot verify · Dormant · Not applicable

WHO DECIDES

Rule engine decides every verdict. Deterministic, no AI.

✓ AI explanation · numbers checked explains, drafts edits, proposes fixes.

Engineer confirms every model edit and every decision.

SAMPLE-DATA CONTRACT · SECTION 5 (CHECKED)

Analyse	96 checks · 2.1 s · 1 fail · 1 warn · 1 ? · 89 pass · 4 dormant
F-01	FD-5-165P · Deck 5 · Fr 165 · P · A-0 fitted, A-60 required
W-01	Deck 5 · Fr 152 · S · 900 mm vs 915 mm (36 in) · draft
U-01	D-5-171-S · no fire class in the model
OR-117	cabin block × 4 · aft of Fr 150 · 8 berths · 14 re-run · 3 changed
F-07	Stair 3B · 1,380 req · 1,350 meas · N 68/44/40/24 · Rev A 1,300
F-08	Deck 5 · Fr 146–150 · P · depth 4.2 m > width 1.2 m
F-09	Deck 5 · Fr 152 · S · 850 mm vs 900 mm
Fixes	1,400 (+20) · N_1 64 → 1,340 (+10) · N_1 62 → 1,320 (+30)
H-01	VD-5-03 · A-60 MVZ bulkhead · Fr 165 · no damper
RCRV	28 POB · D-02-114 30 in < 32 in · D-02-118 34 in

HOW TO READ THIS CANVAS

Each artboard has a hand-off sticky beneath it: purpose,

Tokens

CSS variables in `sonariq.css`. Scope with `.sq.sq-dark` (default) or `.sq.sq-light`. Qt: same names in the QSS palette.

Contrast ratios measured on `--bg-1` · WCAG 2.1 AA

Dark · primary

.sq-dark

--bg-0	--bg-1	--bg-2	--bg-3
#131417	#1b1d21	#24262b	#2d3036
--line	--line-2	--fg-1	--fg-2
#34373e	#4a4e57	#e9eaed 14.0	#b7bbc3 8.8
--fg-3	--accent	--st-fail	--st-fail-fill
#959aa4 6.0	#e392c8 7.4	#ff8a50 7.2	#d55e00 icon
--st-warn	--st-pass	--st-pass-fill	--st-acc
#e69f00 7.5	#3cc49a 7.7	#009e73 icon	#62b8ec 7.7
--st-acc-fill	--st-unk	--st-dor	--fd-ink
#0072b2 icon	#c9cdd4 10.6	#959aa4 6.0	#c9ccd2
--plan-bg	--fc-bg	--fc-bar	--fc-view
#17191c	#2b2b2b host	#323232 host	#3a3d42 host

Light · portal and print

.sq-light

--bg-0	--bg-1	--bg-2	--bg-3
#eceef1	#ffffff	#f5f6f8	#e7e9ed
--line	--line-2	--fg-1	--fg-2
#dcdfe4	#b9bec7	#15171b 17.9	#3d424b 10.1
--fg-3	--accent	--st-fail	--st-fail-fill
#5b616b 6.2	#9e3f7a 6.1	#b34700 5.5	#d55e00 icon
--st-warn	--st-pass	--st-pass-fill	--st-acc
#875a00 6.0	#00785a 5.5	#009e73 icon	#005f96 6.8
--st-acc-fill	--st-unk	--st-dor	--fd-ink
#0072b2 icon	#3d424b 10.1	#5b616b 6.2	#2b2f36
--plan-bg	--st-warn-fill	print --fg-1	print --bg-1
#f7f8fa	#e69f00 icon	#000000 21	#ffffff

TYPE · IBM PLEX SANS (UI) · IBM PLEX MONO, TABULAR FIGURES (NUMBERS, RULE REFS, IDS)

--fs-2xl 24/600	**Stair 3B fails**
--fs-xl 18/600	**Fire & life safety inbox**
--fs-lg 15/600	**Evidence**
portal 14/400	Fire door in the MVZ 4/5 bulkhead
panel 13/400	Corridor clear width below guideline
--fs-sm 12/500	**Deck 5 · Fr 152 · S**
--fs-xs 11/600 caps	REQUIRED VS MEASURED
mono 12 tnum	1,380 mm · 1,350 mm · FD-5-165P · II-2/9.2.2.1.1
mono 15/600	**1,111 · 0,000 · 9,999**

Numbers: thousands separator "," · unit always present · mm default, inches when the rule is written in inches · per-value toggle.

SPACING · 4 PX GRID · RADIUS 4 PX · HAIRLINE 1 PX

--sp-1	4px
--sp-2	8px
--sp-3	12px
--sp-4	16px
--sp-5	20px
--sp-6	24px
--sp-8	32px

--r 4px · every corner

--line 1px · --line-2 for controls

No shadows, blur or gradients anywhere (Qt-buildable).

MOTION

--motion-ping 600ms · ease-out

One ring per changed element, expands once, no repeat. Sonar-ring motif is used only here.

Reduced motion: no ring; the summary count highlights instead.

FIRE-DIVISION LINES · PLAN AND 3D

A-60	———	thick solid · 4 px
A-0	———	medium solid · 2 px
B-15	- - - - - -	dashed · 1.5 px · 6 4
B-0	·····	dotted · 1.5 px · 1.5 3.5

Ink is neutral (--fd-ink). Status colour is never used for division class; a failing division is shown by the highlight + pin, not by recolouring the line style.

SELECTION VS STATUS

-- accent	selection, focus ring, active tab, ping. Never a verdict.
✕ ⚠ ✓ ⚑ ? ○ –	status: shape + glyph + word

SONARIQ.CSS · TOKEN EXCERPT (FULL FILE IN PROJECT/SONARIQ.CSS)

```
.sq, .sq-dark {
  --bg-0: #131417;  --bg-1: #1b1d21;
  --bg-2: #24262b;  --bg-3: #2d3036;
  --line: #34373e;  --line-2: #4a4e57;
  --fg-1: #e9eaed;  --fg-2: #b7bbc3;
  --fg-3: #959aa4;  --fg-inv: #131417;
  --accent: #e392c8;
  --st-fail: #ff8a50; --st-fail-fill: #d55e00;
  --st-warn: #e69f00; --st-warn-fill: #e69f00;
  --st-pass: #3cc49a; --st-pass-fill: #009e73;
  --st-acc:  #62b8ec; --st-acc-fill:  #0072b2;
  --st-unk:  #c9cdd4;  /* dashed outline */
  --st-dor:  #959aa4;  --st-na: #959aa4;
  --fd-ink:  #c9ccd2;
}
.sq-light {
  --bg-0: #eceef1;  --bg-1: #ffffff;
  --bg-2: #f5f6f8;  --bg-3: #e7e9ed;
  --line: #dcdfe4;  --line-2: #b9bec7;
  --fg-1: #15171b;  --fg-2: #3d424b;
  --fg-3: #5b616b;  --fg-inv: #ffffff;
  --accent: #9e3f7a;
  --st-fail: #b34700; --st-warn: #875a00;
  --st-pass: #00785a; --st-acc:  #005f96;
  --st-unk:  #3d424b; --st-dor:  #5b616b;
  --fd-ink:  #2b2f36;
}
.sq {
  --font-ui: "IBM Plex Sans", system-ui;
  --font-mono: "IBM Plex Mono", monospace;
  --fs-panel: 13px;  --fs-portal: 14px;
  --fs-xs: 11px; --fs-sm: 12px; --fs-md: 13px;
  --fs-lg: 15px; --fs-xl: 18px; --fs-2xl: 24px;
  --sp-1: 4px; --sp-2: 8px; --sp-3: 12px;
  --sp-4: 16px; --sp-6: 24px; --sp-8: 32px;
  --r: 4px;  --hair: 1px;
  --fd-a60-w: 4px;  --fd-a0-w: 2px;
  --fd-b15-w: 1.5px; --fd-b15-dash: 6 4;
  --fd-b0-w: 1.5px;  --fd-b0-dash: 1.5 3.5;
```